

DBS Chapter 6 – Basic SQL

SQL Basics

1. What is SQL?

SQL (Structured Query Language) is a standard language used to create, manage, and manipulate relational databases.

2. Why is SQL called a comprehensive database language?

Because it supports data definition (DDL), data manipulation (DML), constraints, security, views, and transaction control.

3. Difference between DDL and DML?

DDL defines database structure (CREATE, ALTER, DROP). DML manipulates data (SELECT, INSERT, UPDATE, DELETE).

4. What is a schema in SQL?

A schema is a logical collection of tables, views, constraints, and other database objects owned by a user.

5. What is a catalog?

A catalog contains information about all schemas in the database.

Constraints

6. Explain NOT NULL.

Ensures that an attribute must have a value and cannot contain NULL.

7. Explain DEFAULT.

Assigns a default value if no explicit value is provided during insertion.

8. Explain CHECK constraint.

Restricts attribute values based on a condition.

9. Explain PRIMARY KEY.

A unique and NOT NULL identifier for each tuple.

10. UNIQUE vs PRIMARY KEY.

Both enforce uniqueness, but PRIMARY KEY cannot contain NULL and only one exists per table.

11. Explain FOREIGN KEY.

Ensures referential integrity by referencing a primary key in another table.

12. Referential actions.

Options include RESTRICT, CASCADE, SET NULL, and SET DEFAULT.

Queries & Operations

13. Explain SELECT-FROM-WHERE.

SELECT chooses attributes, FROM specifies tables, WHERE filters rows.

14. What is projection?

Selecting specific columns in SELECT clause.

15. What is selection?

Filtering rows using WHERE clause.

16. What is JOIN?

Combines rows from multiple tables based on a related column.

17. What is aliasing?

Renaming tables or columns for readability.

18. What happens if WHERE is missing?

All rows are returned. If multiple tables exist, cross product occurs.

19. UNION vs UNION ALL.

UNION removes duplicates; UNION ALL keeps duplicates.

20. What is INTERSECT?

Returns common rows between queries.

21. What is EXCEPT?

Returns rows from first query not in second.

22. LIKE operator.

Used for pattern matching with % and _ wildcards.

23. BETWEEN operator.

Checks if value lies within a range.

24. ORDER BY.

Sorts result set ascending or descending.

25. INSERT statement.

Adds new rows into a table.

26. DELETE statement.

Removes rows from a table.

27. UPDATE statement.

Modifies existing rows.

Advanced Topics

28. Nested queries.

Queries inside another query.

29. Aggregate functions.

Functions like COUNT, SUM, AVG, MIN, MAX.

30. GROUP BY.

Groups rows for aggregate calculations.

31. Views.

Virtual tables created using CREATE VIEW.

32. Triggers.

Automatically executed actions when events occur.

33. GRANT & REVOKE.

Used to give or remove user privileges.

34. Transactions.

Logical unit of work controlled by COMMIT and ROLLBACK.

Mega List of Example SQL Queries

```
CREATE TABLE Student (ID INT PRIMARY KEY, Name VARCHAR(50), Age INT CHECK (Age > 0));
ALTER TABLE Student ADD Email VARCHAR(100);
INSERT INTO Student VALUES (1, 'Ali', 20);
INSERT INTO Student (ID, Name) VALUES (2, 'Sara');
INSERT INTO Student (ID, Name, Age) SELECT ID, Name, Age FROM OldStudent;
SELECT * FROM Student;
SELECT Name, Age FROM Student WHERE Age > 18;
SELECT * FROM Student WHERE Name LIKE 'A%';
SELECT * FROM Student WHERE Age BETWEEN 18 AND 25;
SELECT DISTINCT Age FROM Student;
SELECT S.Name, D.DeptName FROM Student S, Department D WHERE S.DeptID = D.ID;
SELECT Name FROM Student UNION SELECT Name FROM Alumni;
SELECT Name FROM Student INTERSECT SELECT Name FROM Alumni;
SELECT Name FROM Student EXCEPT SELECT Name FROM Alumni;
UPDATE Student SET Age = 21 WHERE ID = 1;
DELETE FROM Student WHERE ID = 2;
SELECT COUNT(*), AVG(Age) FROM Student;
SELECT DeptID, COUNT(*) FROM Student GROUP BY DeptID;
CREATE VIEW AdultStudents AS SELECT * FROM Student WHERE Age >= 18;
CREATE TRIGGER AgeCheck BEFORE INSERT ON Student FOR EACH ROW BEGIN IF NEW.Age < 0 THEN SIGNAL SQLSTATE '45000'; END IF; END;
GRANT SELECT, INSERT ON Student TO User1;
REVOKE INSERT ON Student FROM User1;
COMMIT;
ROLLBACK;
```